

TFM - Stacked y -maps and μy -maps

Joan Ribot Oliver

2026

Universidad de Cantabria

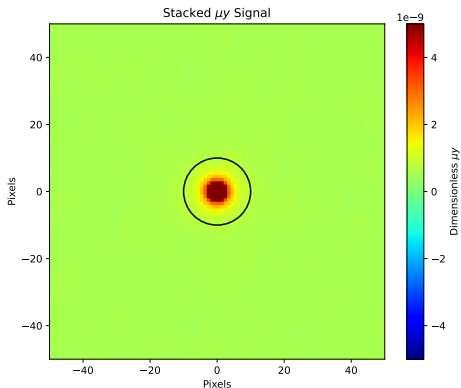
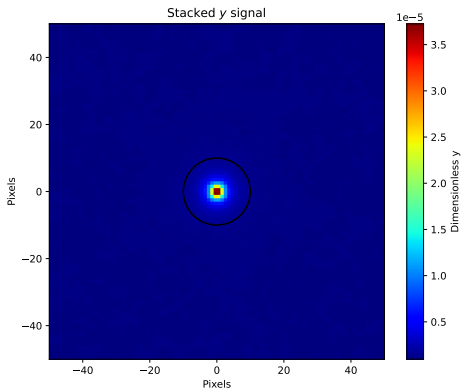
Input stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 5 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

Input Data stacked patch of 1000 brightest clusters, $\mu = 5\text{e-}04$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10\text{p}$



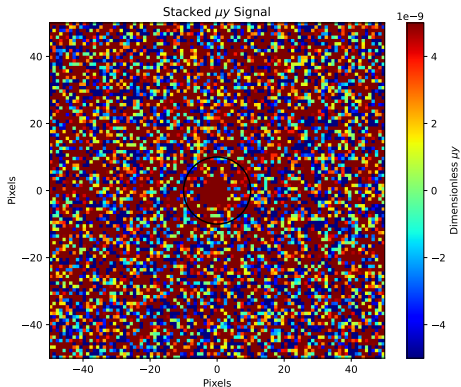
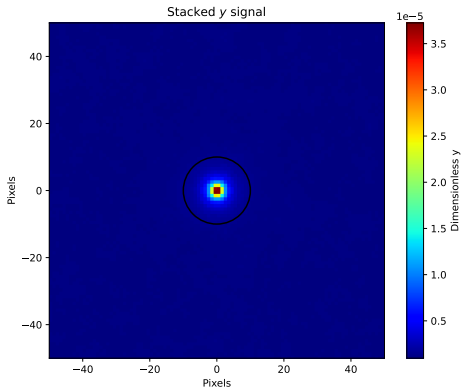
ILC stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 5 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

P-ILC stacked patch of 1000 brightest clusters, $\mu = 5\text{e-}04$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10\text{p}$



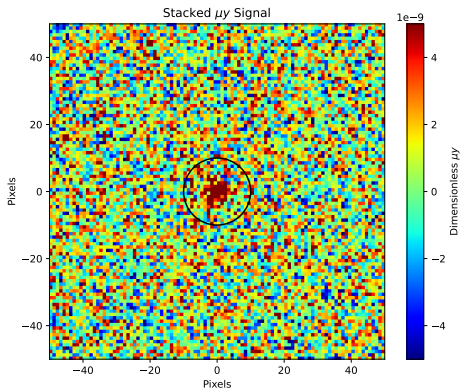
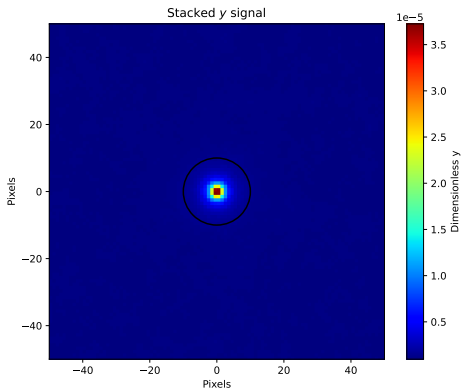
CILC stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 5 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

P-CILC stacked patch of 1000 brightest clusters, $\mu = 5\text{e-}04$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10\text{p}$



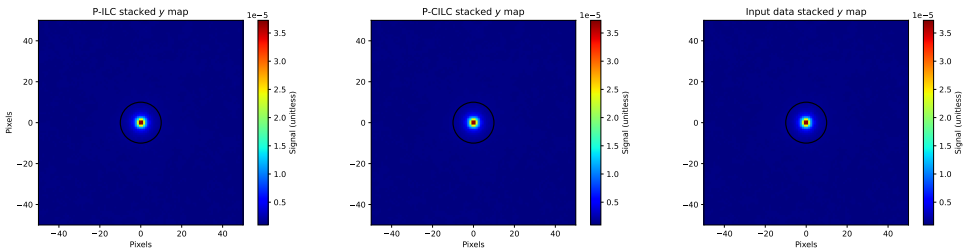
Comparison γ stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 5 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

Comparison of γ map stacked patches of 1000 brightest clusters, μ value = $5e-04$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10p$



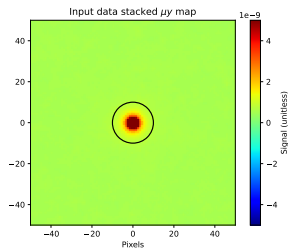
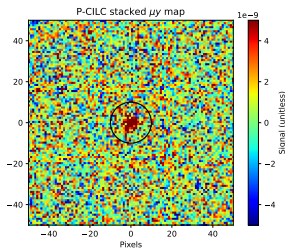
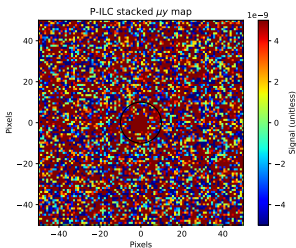
Comparison μy stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 5 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

Comparison of μy map stacked patches of 1000 brightest clusters, μ value = $5e-04$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10p$



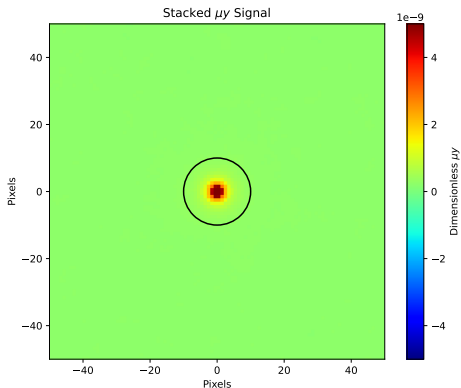
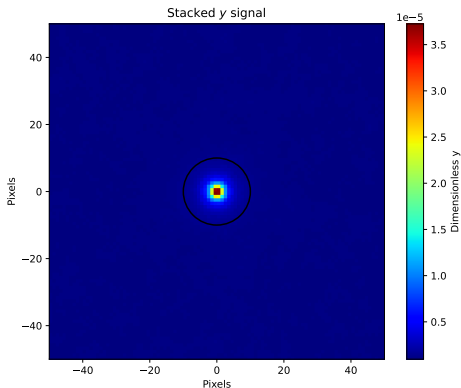
Input stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

Input Data stacked patch of 1000 brightest clusters, $\mu = 2\text{e-}04$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10\text{p}$



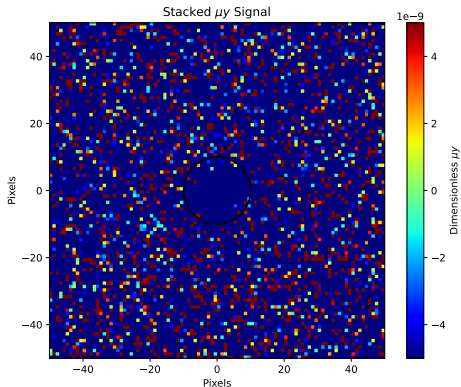
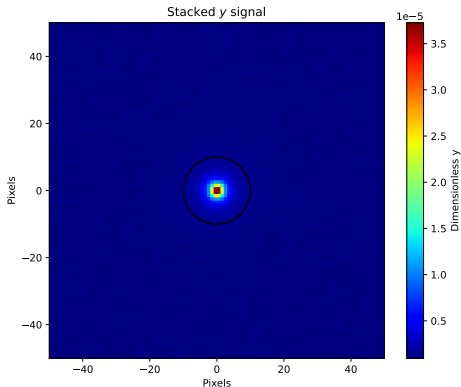
ILC stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

P-ILC stacked patch of 1000 brightest clusters, $\mu = 2\text{e-}04$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10\text{p}$



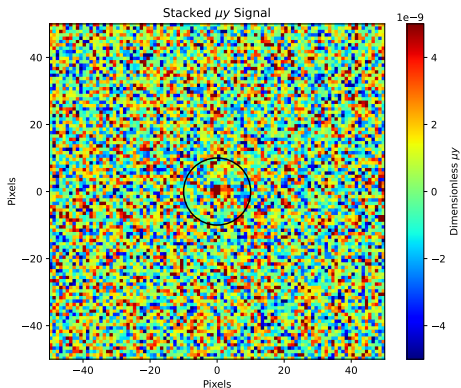
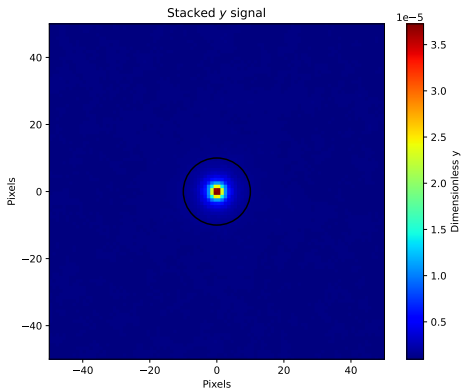
CILC stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

P-CILC stacked patch of 1000 brightest clusters, $\mu = 2\text{e-}04$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10\text{p}$



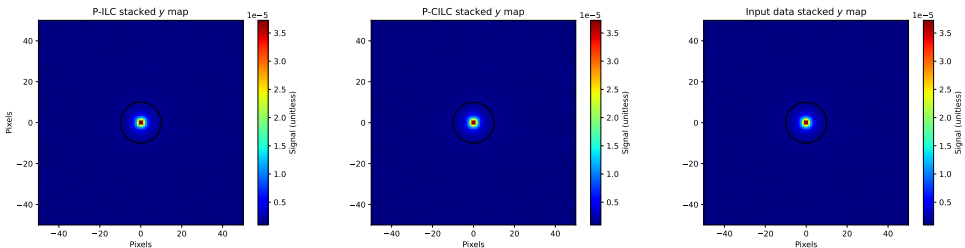
Comparison y stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

Comparison of y map stacked patches of $1e+03$ brightest clusters, μ value = $2e-04$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10p$



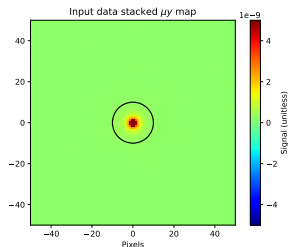
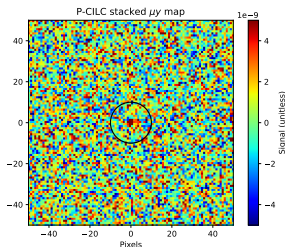
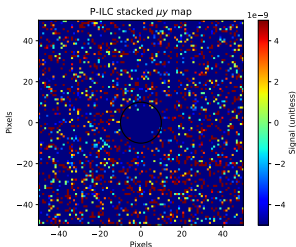
Comparison $\mu\gamma$ stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

Comparison of $\mu\gamma$ map stacked patches of $1e+03$ brightest clusters, μ value = $2e-04$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10p$



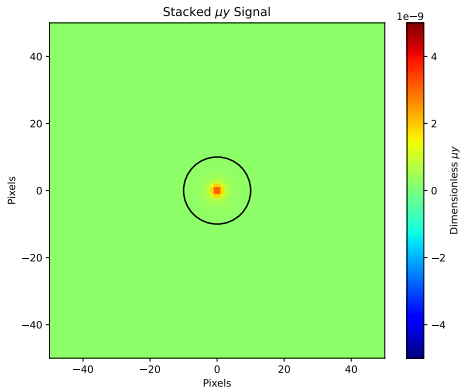
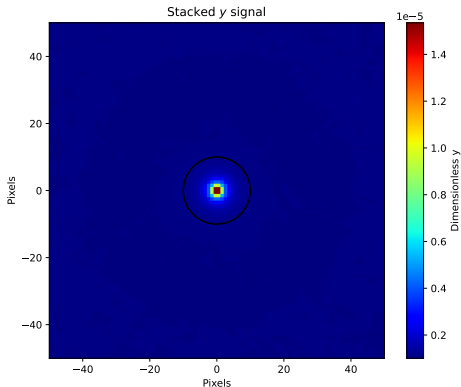
Input stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^4

Input Data stacked patch of 10000 brightest clusters, $\mu = 2\text{e-}04$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10$



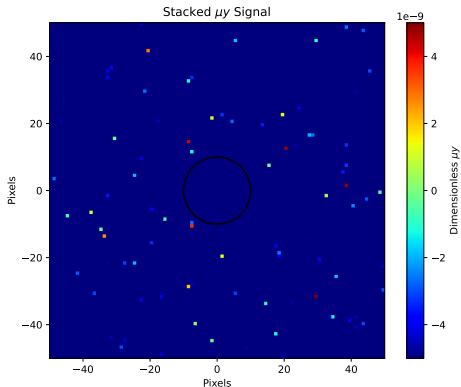
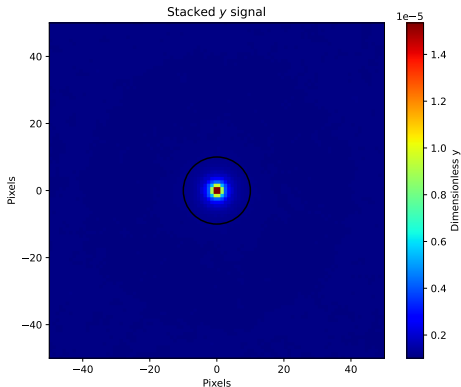
ILC stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^4

P-ILC stacked patch of 10000 brightest clusters, $\mu = 2\text{e-}04$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10\text{p}$



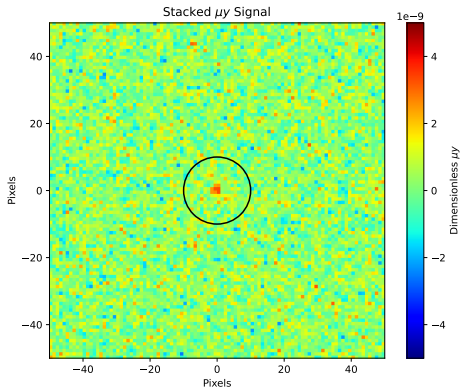
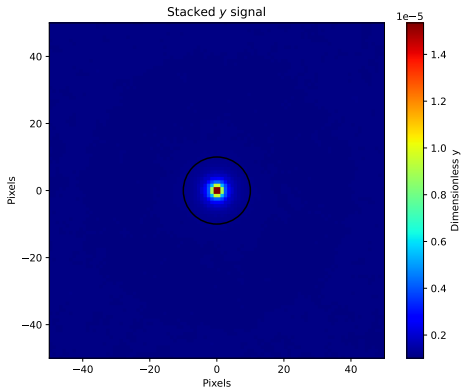
CILC stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^4

P-CILC stacked patch of 10000 brightest clusters, $\mu = 2\text{e-}04$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10\text{p}$



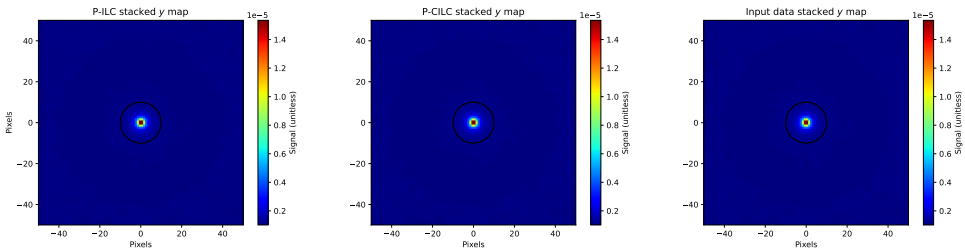
Comparison y stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^4

Comparison of y map stacked patches of $1e+04$ brightest clusters, μ value = $2e-04$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10p$



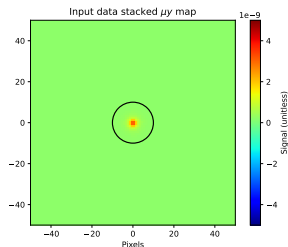
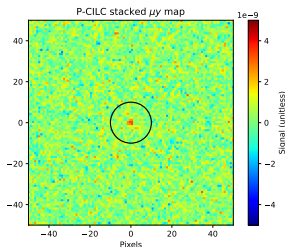
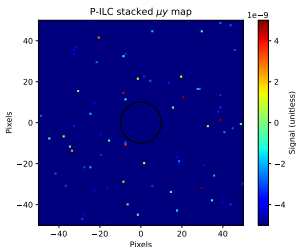
Comparison $\mu\gamma$ stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^4

Comparison of $\mu\gamma$ map stacked patches of $1e+04$ brightest clusters, μ value = $2e-04$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10p$



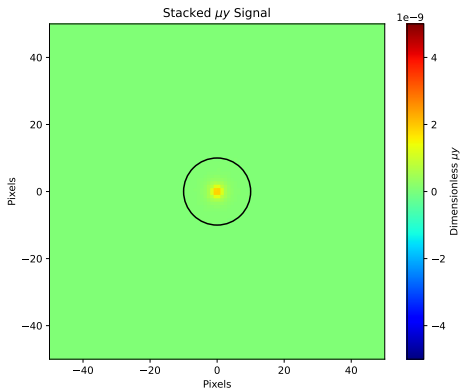
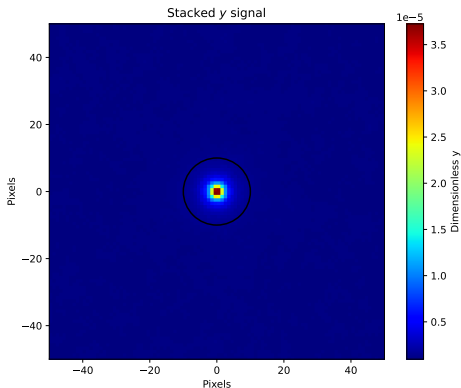
Input stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 5 \times 10^{-5}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

Input Data stacked patch of 1000 brightest clusters, $\mu = 5\text{e-}05$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10$ p



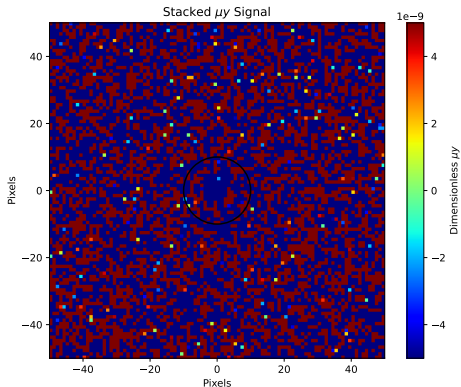
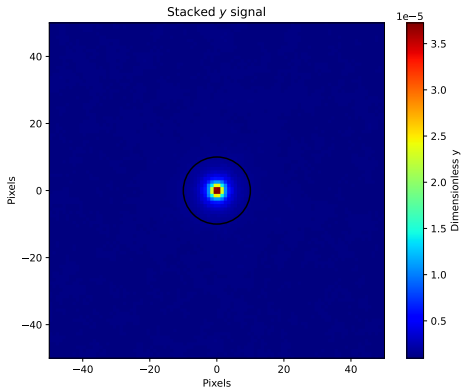
ILC stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 5 \times 10^{-5}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

P-ILC stacked patch of 1000 brightest clusters, $\mu = 5\text{e-}05$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10\text{p}$



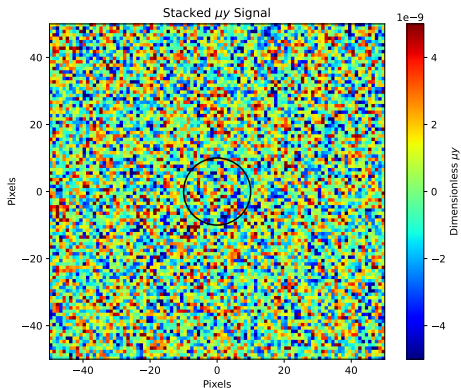
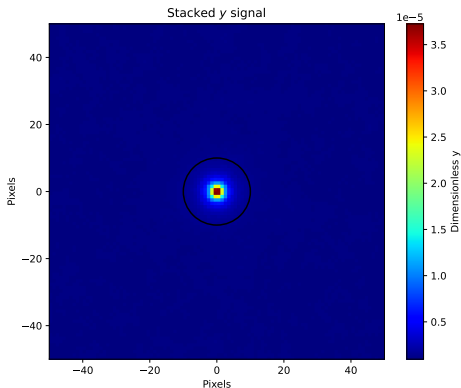
CILC stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 5 \times 10^{-5}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

P-CILC stacked patch of 1000 brightest clusters, $\mu = 5e-05$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10p$



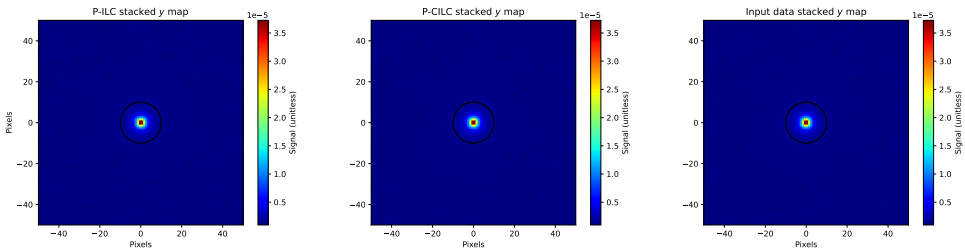
Comparison y stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 5 \times 10^{-5}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

Comparison of y map stacked patches of $1e+03$ brightest clusters, μ value = $5e-05$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10p$



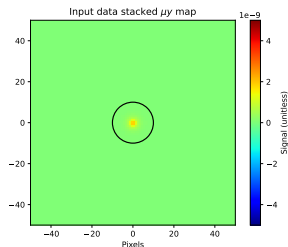
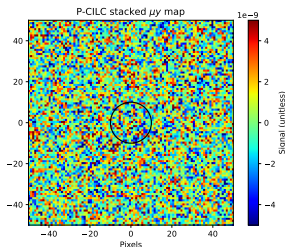
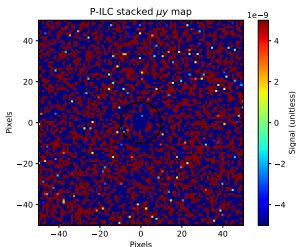
Comparison $\mu\gamma$ stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 5 \times 10^{-5}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

Comparison of $\mu\gamma$ map stacked patches of $1e+03$ brightest clusters, μ value = $5e-05$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10p$



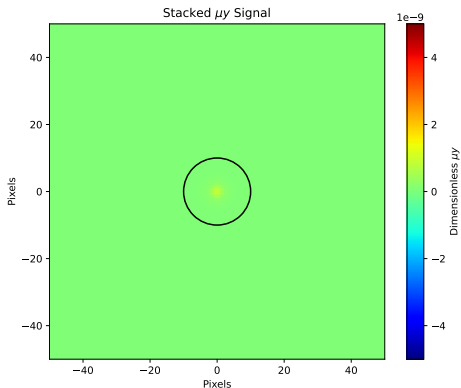
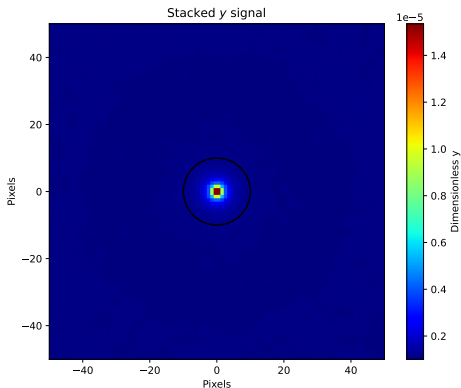
Input stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 5 \times 10^{-5}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^4

Input Data stacked patch of 10000 brightest clusters, $\mu = 5\text{e-}05$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10$



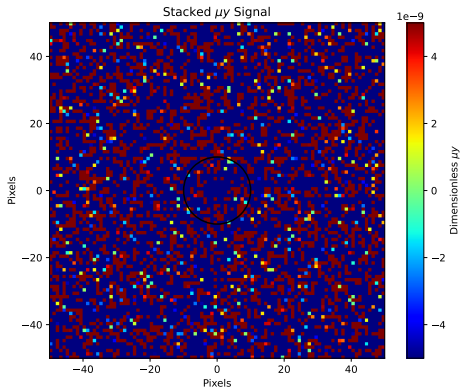
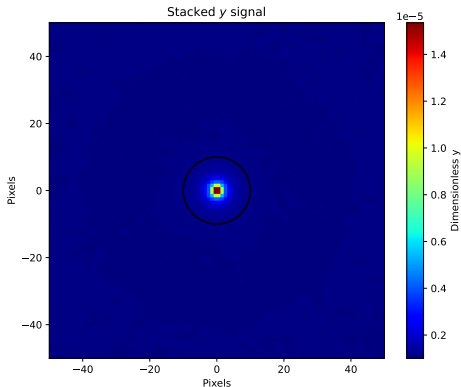
ILC stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 5 \times 10^{-5}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^4

P-ILC stacked patch of 10000 brightest clusters, $\mu = 5\text{e-}05$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10\text{p}$



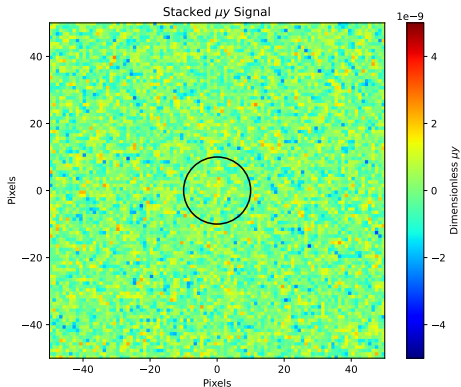
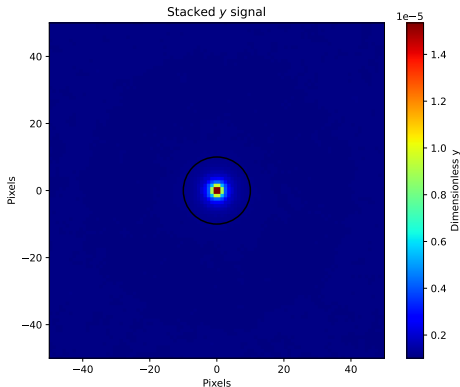
CILC stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 5 \times 10^{-5}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^4

P-CILC stacked patch of 10000 brightest clusters, $\mu = 5\text{e-}05$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10\text{p}$



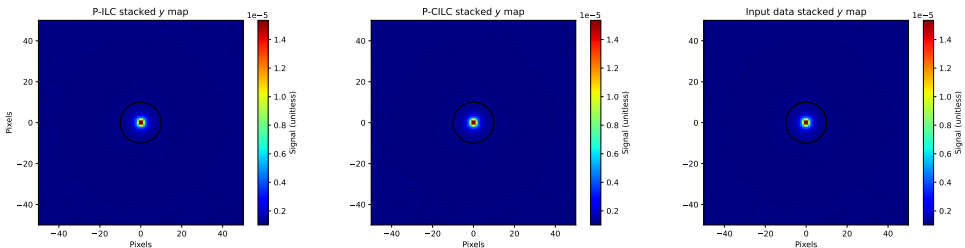
Comparison y stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 5 \times 10^{-5}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^4

Comparison of y map stacked patches of $1e+04$ brightest clusters, μ value = $5e-05$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10p$



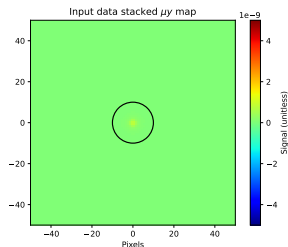
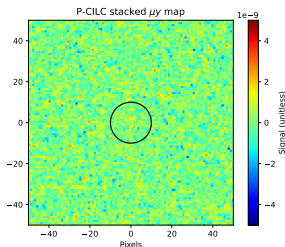
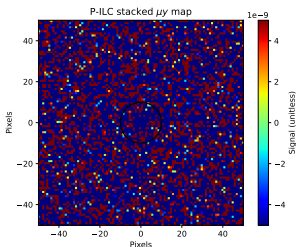
Comparison $\mu\gamma$ stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 5 \times 10^{-5}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^4

Comparison of $\mu\gamma$ map stacked patches of $1e+04$ brightest clusters, μ value = $5e-05$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10p$



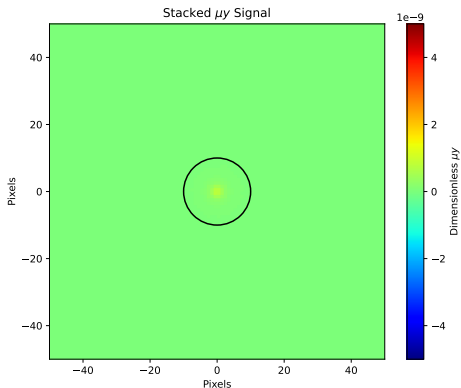
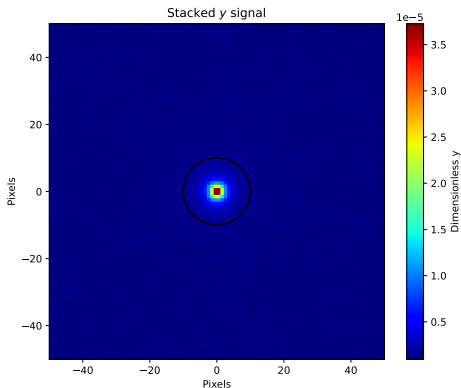
Input stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 2 \times 10^{-5}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

Input Data stacked patch of 1000 brightest clusters, $\mu = 2e-05$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10$ p



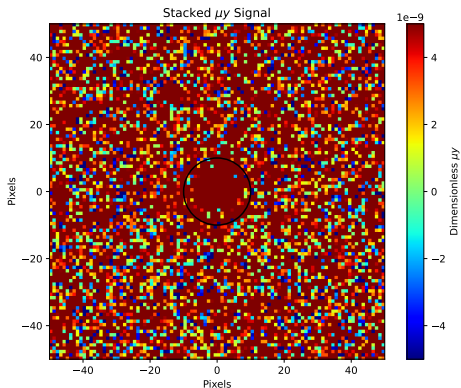
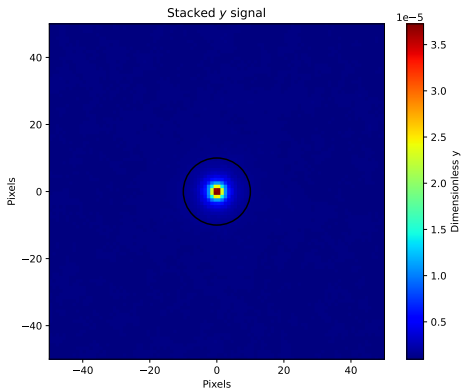
ILC stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 2 \times 10^{-5}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

P-ILC stacked patch of 1000 brightest clusters, $\mu = 2\text{e-}05$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10\text{p}$



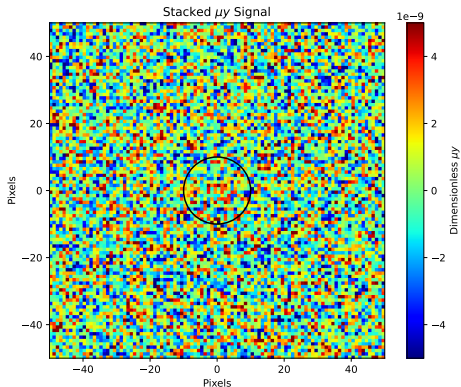
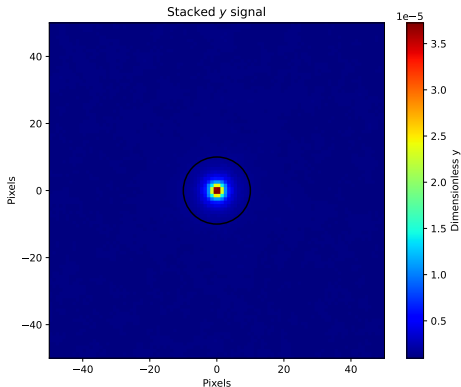
CILC stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 2 \times 10^{-5}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

P-CILC stacked patch of 1000 brightest clusters, $\mu = 2\text{e-}05$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10\text{p}$



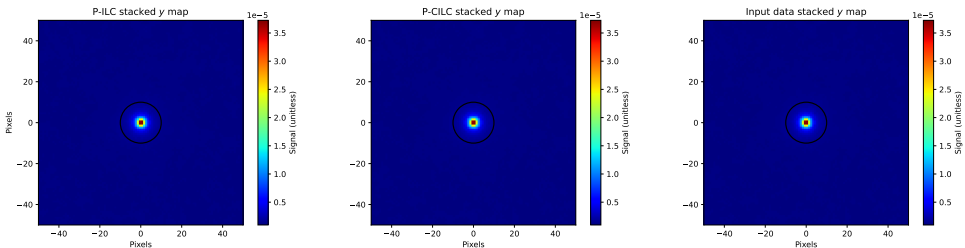
Comparison y stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 2 \times 10^{-5}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

Comparison of y map stacked patches of $1e+03$ brightest clusters, μ value = $2e-05$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10p$



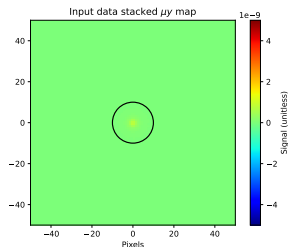
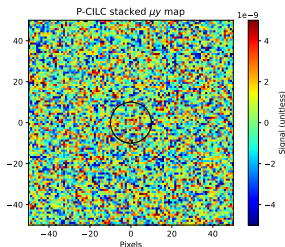
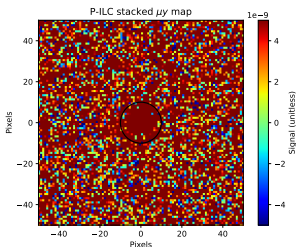
Comparison $\mu\gamma$ stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 2 \times 10^{-5}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

Comparison of $\mu\gamma$ map stacked patches of $1e+03$ brightest clusters, μ value = $2e-05$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10p$



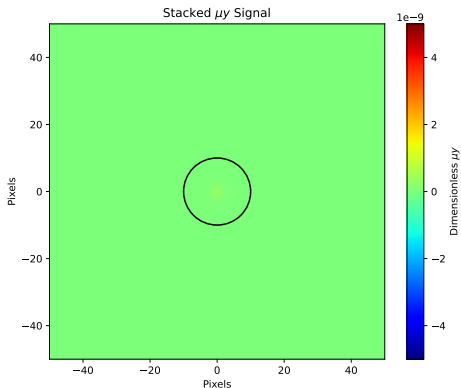
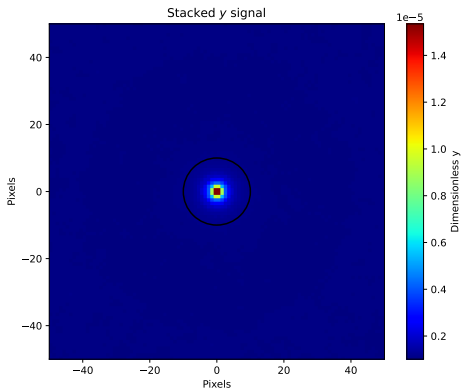
Input stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 2 \times 10^{-5}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^4

Input Data stacked patch of 10000 brightest clusters, $\mu = 2\text{e-}05$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10$



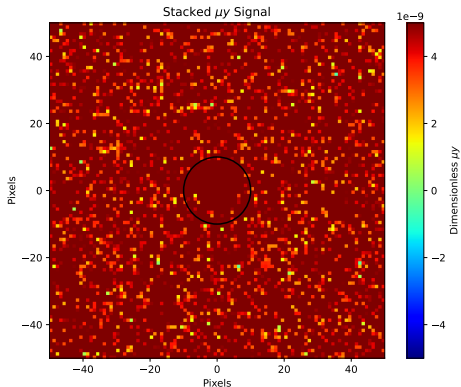
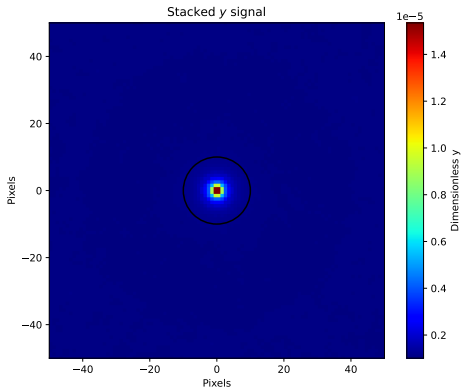
ILC stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 2 \times 10^{-5}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^4

P-ILC stacked patch of 10000 brightest clusters, $\mu = 2e-05$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10p$



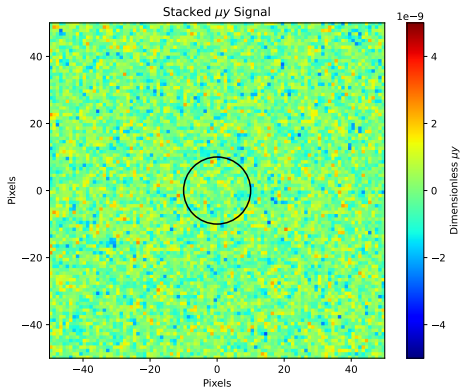
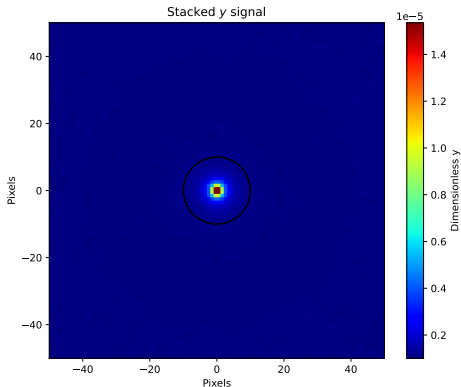
CILC stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 2 \times 10^{-5}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^4

P-CILC stacked patch of 10000 brightest clusters, $\mu = 2\text{e-}05$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10\text{p}$



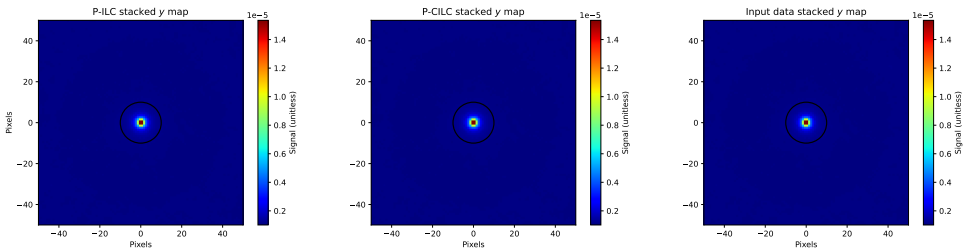
Comparison y stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 2 \times 10^{-5}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^4

Comparison of y map stacked patches of $1e+04$ brightest clusters, μ value = $2e-05$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10p$



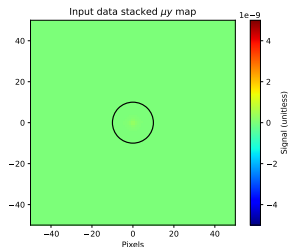
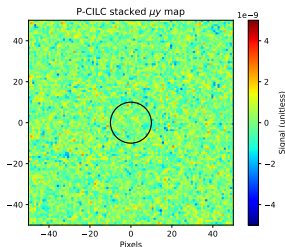
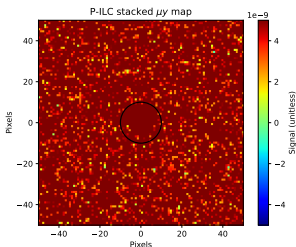
Comparison $\mu\gamma$ stacked zoomed maps

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

$\mu = 2 \times 10^{-5}$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^4

Comparison of $\mu\gamma$ map stacked patches of $1e+04$ brightest clusters, μ value = $2e-05$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10p$



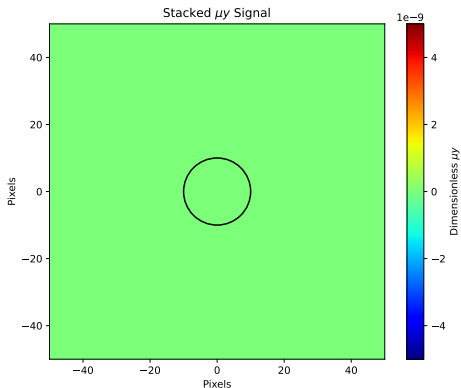
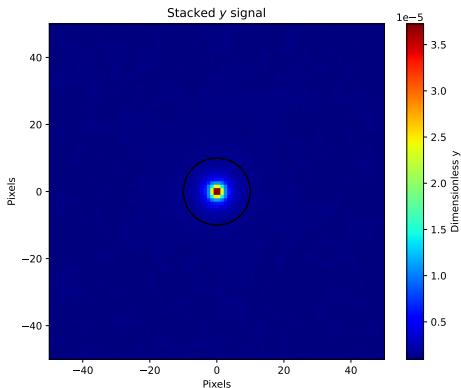
Input stacked zoomed maps

3 layers: tSZ, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

Null test: $\mu = 0$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

Input Data stacked patch of 1000 brightest clusters, $\mu = 0e+00$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10p$



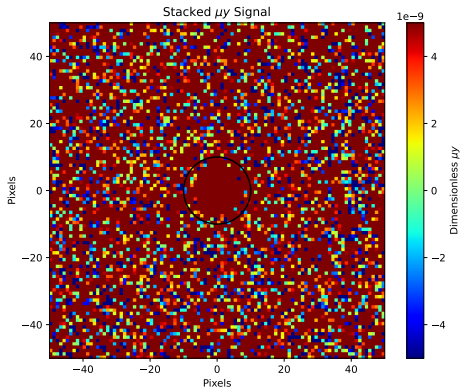
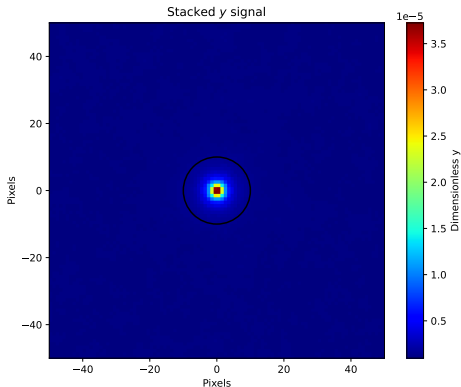
ILC stacked zoomed maps

3 layers: tSZ, CMB and noise ($0.2\mu\text{K}\cdot\text{arcmin}$)

Null test: $\mu = 0$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

P-ILC stacked patch of 1000 brightest clusters, $\mu = 0\text{e}+00$, base sensitivity = $0.20\mu\text{K}\cdot\text{arcmin}$, $r = 10\text{p}$



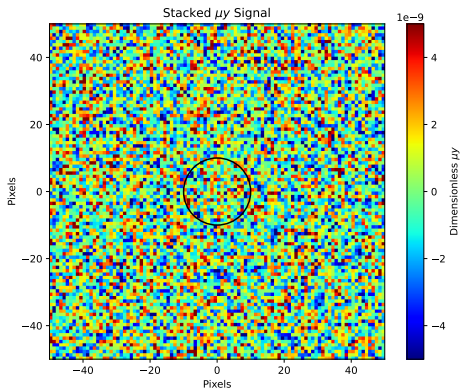
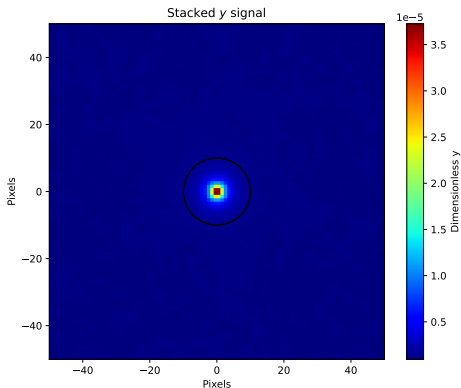
CILC stacked zoomed maps

3 layers: tSZ, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

Null test: $\mu = 0$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

P-CILC stacked patch of 1000 brightest clusters, $\mu = 0e+00$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10p$



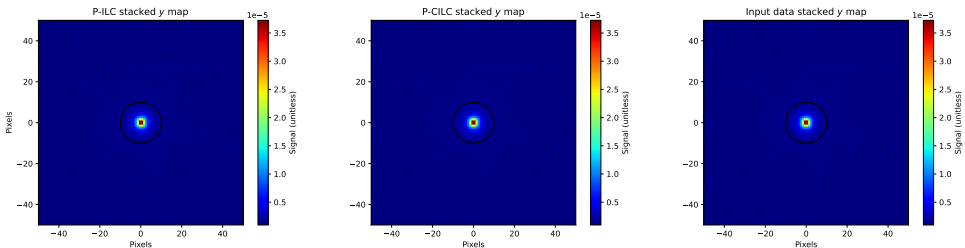
Comparison y stacked zoomed maps

3 layers: tSZ, CMB and noise ($0.2\mu\text{K}\cdot\text{arcmin}$)

Null test: $\mu = 0$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

Comparison of y map stacked patches of $1e+03$ brightest clusters, μ value = $0e+00$, base sensitivity = $0.20\mu\text{K}\cdot\text{arcmin}$, $r = 10p$



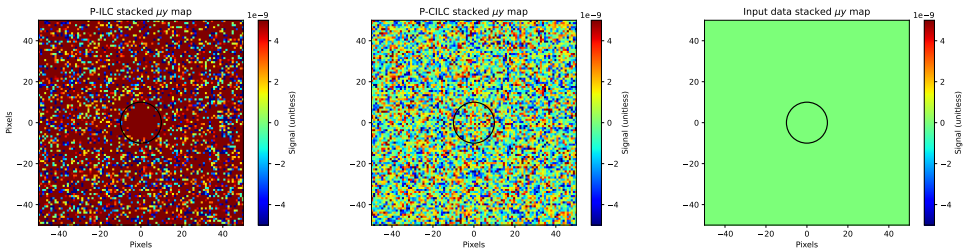
Comparison μy stacked zoomed maps

3 layers: tSZ, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

Null test: $\mu = 0$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^3

Comparison of μy map stacked patches of $1e+03$ brightest clusters, μ value = $0e+00$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10p$



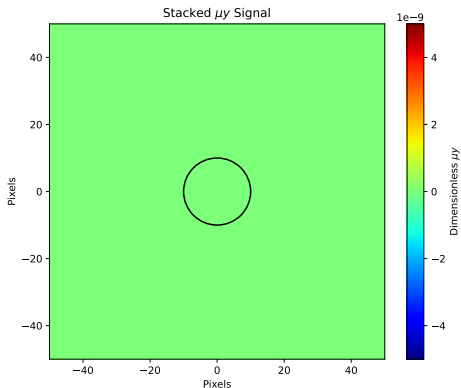
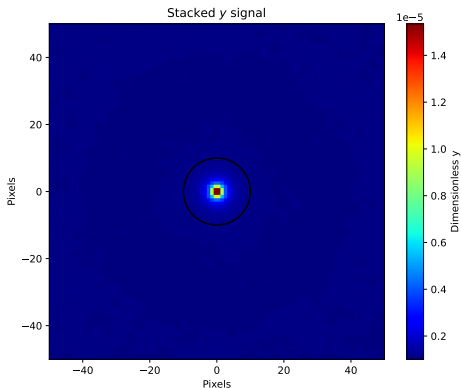
Input stacked zoomed maps

3 layers: tSZ, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

Null test: $\mu = 0$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^4

Input Data stacked patch of 10000 brightest clusters, $\mu = 0e+00$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10p$



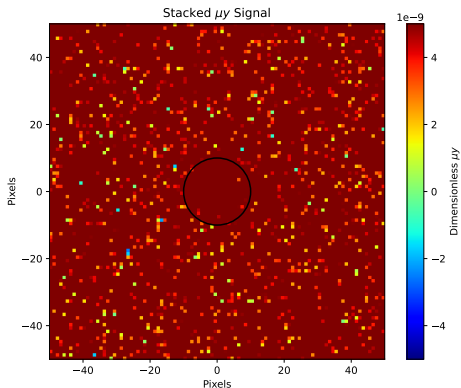
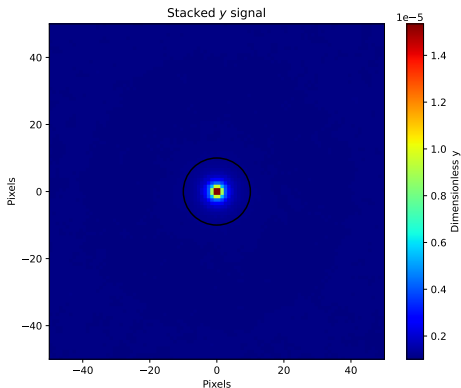
ILC stacked zoomed maps

3 layers: tSZ, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

Null test: $\mu = 0$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^4

P-ILC stacked patch of 10000 brightest clusters, $\mu = 0e+00$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10p$



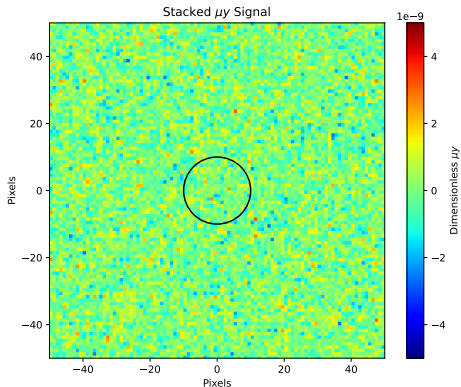
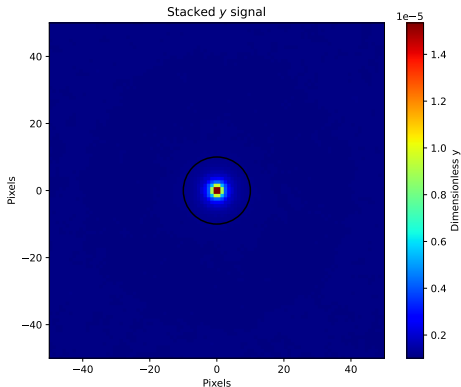
CILC stacked zoomed maps

3 layers: tSZ, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

Null test: $\mu = 0$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^4

P-CILC stacked patch of 10000 brightest clusters, $\mu = 0e+00$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10p$



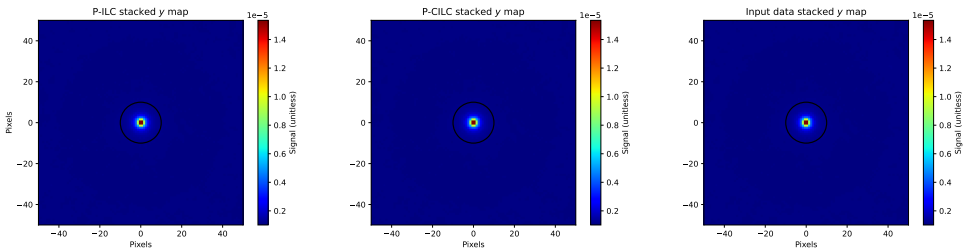
Comparison y stacked zoomed maps

3 layers: tSZ, CMB and noise ($0.2\mu\text{K}\cdot\text{arcmin}$)

Null test: $\mu = 0$, freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$ pixels, stacked maps = 10^4

Comparison of y map stacked patches of $1e+04$ brightest clusters, μ value = $0e+00$, base sensitivity = $0.20\mu\text{K}\cdot\text{arcmin}$, $r = 10p$



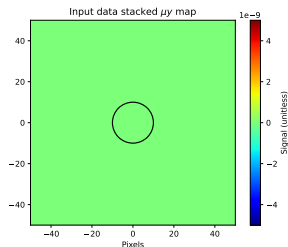
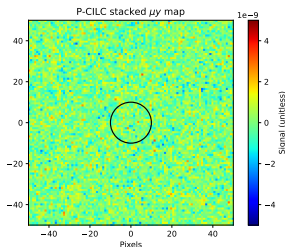
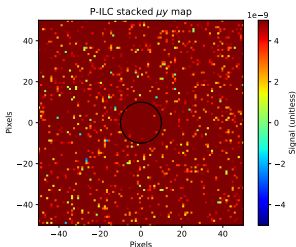
Comparison $\mu\gamma$ stacked zoomed maps

3 layers: tSZ, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

Null test: $\mu = 0$, freqs = [30, 40, 90, 150, 220, 270] GHz

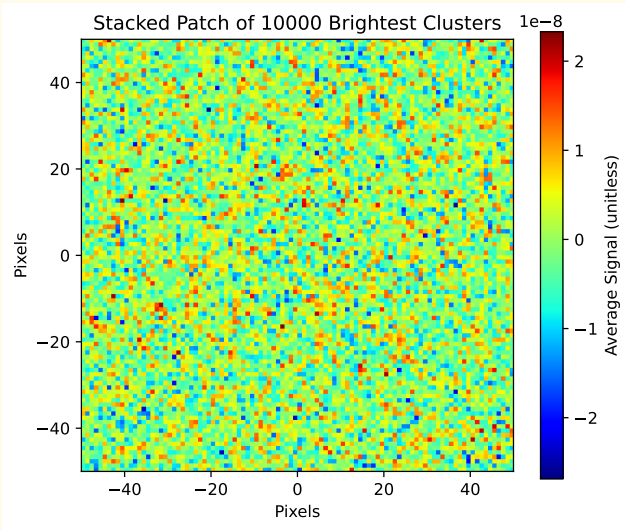
$r = 10$ pixels, stacked maps = 10^4

Comparison of $\mu\gamma$ map stacked patches of $1e+04$ brightest clusters, μ value = $0e+00$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r = 10p$



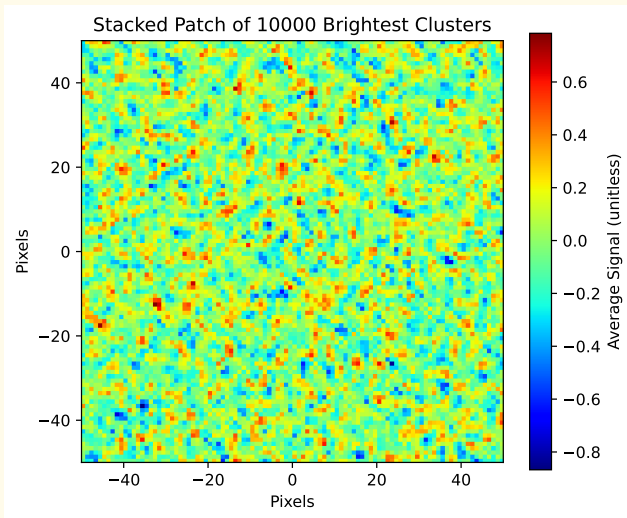
P-ILC stacked zoomed maps (no y input)

3 layers: μ -distortion, CMB and noise ($30\mu K \cdot \text{arcmin}$)
 $\mu = 2 \times 10^{-8}$, freqs = [30, 40, 90, 150, 220, 270] GHz



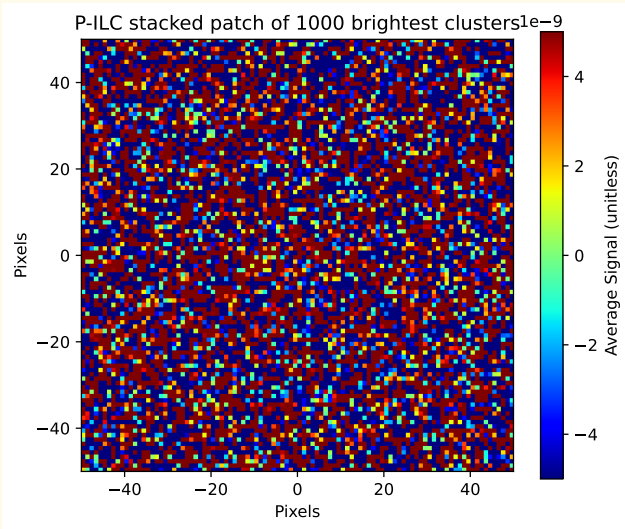
H-ILC stacked zoomed maps (no y input)

3 layers: μ -distortion, CMB and noise ($30\mu K \cdot \text{arcmin}$)
 $\mu = 2 \times 10^{-8}$, freqs = [30, 40, 90, 150, 220, 270] GHz



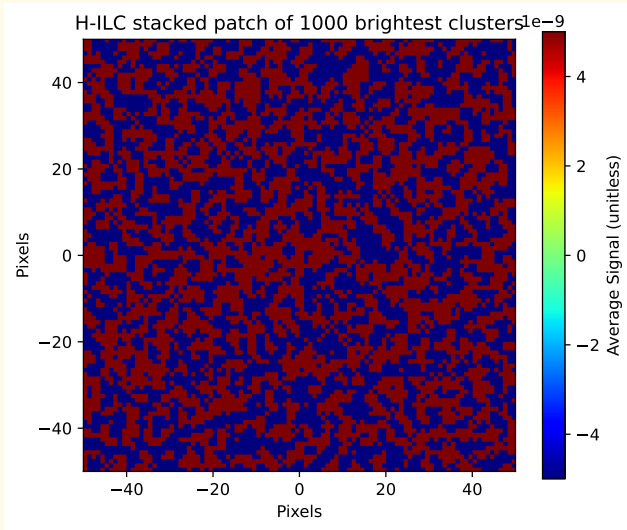
P-ILC stacked zoomed maps (no y input)

3 layers: μ -distortion, CMB and noise ($2\mu K \cdot \text{arcmin}$)
 $\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz



H-ILC stacked zoomed maps (no y input)

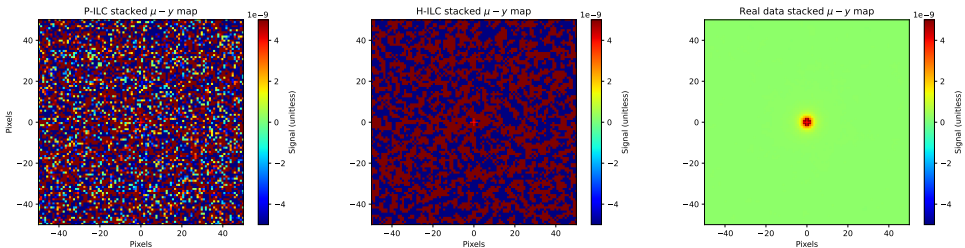
3 layers: μ -distortion, CMB and noise ($2\mu K \cdot \text{arcmin}$)
 $\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz



Comparison of stacked zoomed maps (no y input)

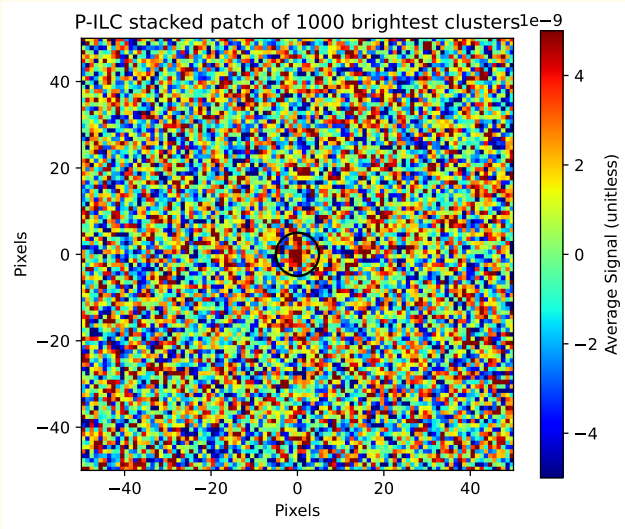
3 layers: μ -distortion, CMB and noise ($2\mu K \cdot \text{arcmin}$)
 $\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz

Comparison of stacked patches of 1000 brightest clusters, μ value = $2e-04$



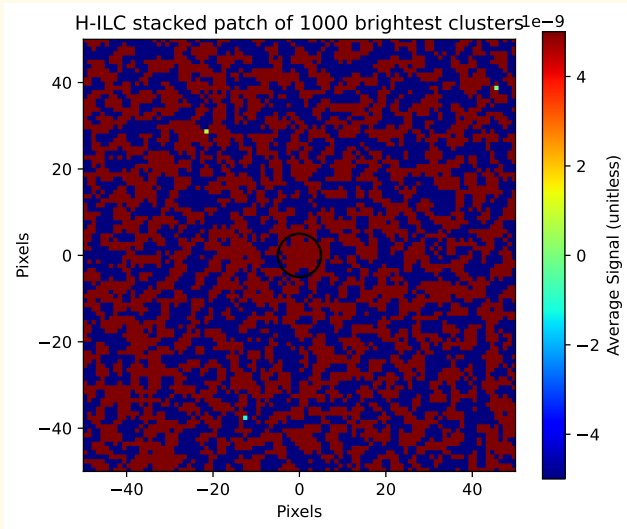
P-ILC stacked zoomed maps (no y input)

3 layers: μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)
 $\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz



H-ILC stacked zoomed maps (no y input)

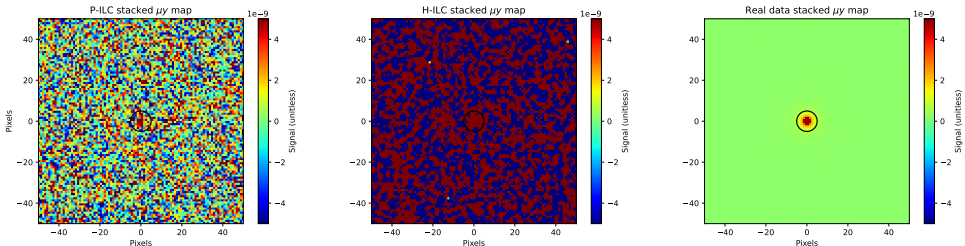
3 layers: μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)
 $\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz



Comparison of stacked zoomed maps (no y input)

3 layers: μ -distortion, CMB and noise ($0.2\mu\text{K}\cdot\text{arcmin}$)
 $\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz

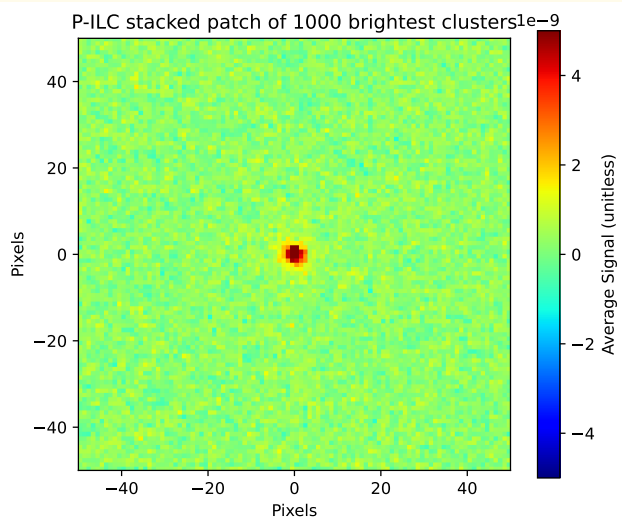
Comparison of stacked patches of 1000 brightest clusters, μ value = $2\text{e-}04$, base sensitivity = $0.20\mu\text{K}\cdot\text{arcmin}$



P-ILC stacked zoomed maps (no y input)

3 layers: μ -distortion, CMB and noise ($0.02\mu K \cdot \text{arcmin}$)

$\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz



H-ILC stacked zoomed maps (no y input)

3 layers: μ -distortion, CMB and noise ($0.02\mu K \cdot \text{arcmin}$)

$\mu = 2 \times 10^{-4}$, freqs = [30, 40, 90, 150, 220, 270] GHz

